REVISION 2021 • SEMESTER 2 • ENGINEERING SCIENCE LAB

Basic CAD Lab

A complete practical handbook for Course 2028. It converts the official Basic CAD Lab syllabus into a workshop-style manual for learning, CAD record preparation, lab examination and real drafting practice.

COURSE CODE
2028

COURSE TITLE
Basic CAD Lab

SEMESTER
2

CREDITS
No Credit

PROGRAMMES
Mechanical / Tool and Die / Manufacturing Technology

PERIODS/WEEK
3 (L:0 T:0 P:3)

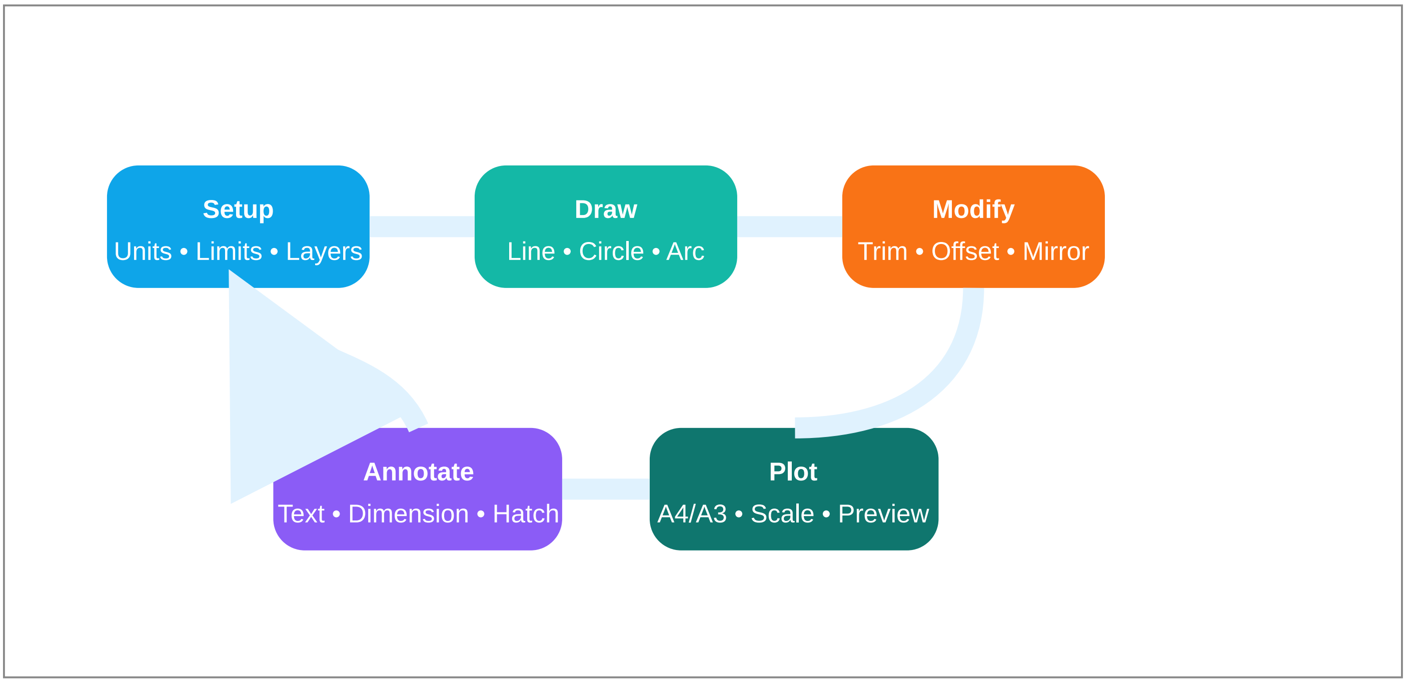
PERIODS/SEMESTER
45

CATEGORY
Engineering Science

> CAD Lab-▯ mark ▯▯▯▯▯ command names ▯▯▯▯▯▯ ▯▯▯▯. Drawing setup, layer control, object snap, dimensions, plotting, record printout ▯▯▯▯▯ practical steps ▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯ practice ▯▯▯▯.

CAD drafting workflow

Industrial drawing is not just a picture. It is an instruction document for manufacturing, inspection and assembly.



Course overview

The supplied syllabus defines Basic CAD Lab as a 45-hour practical course for introducing computer-aided drafting software and 2D drawing/design/drafting using AutoCAD-type tools.

Why it matters

CAD drawing is used in manufacturing, tool design, machine shop drawings, inspection sheets, layout drawings and fabrication communication. A wrong line, dimension, layer or plot scale can create a wrong component.

Prerequisite

Engineering Graphics knowledge: projections, orthographic views, sectioning, dimensions and basic drawing conventions.

Assessment reality

This is not a theory memory subject. End-semester examination is CAD practical only. Record book with printed/plotted drawings is required.

Course outcomes and engineering meaning

CO	Official outcome	Duration	Meaning in lab
CO1	Illustrate the use of CAD software.	6 h	Open software, identify window parts, understand applications, start/save drawings.
CO2	Identify various commands used in CAD.	9 h	Use draw, modify, coordinate, dimension, text and plot commands correctly.
CO3	Apply knowledge to draw simple two-dimensional drawings and sections using CAD.	21 h	Produce neat orthographic and sectional drawings with layers, hatch and dimensions.
CO4	Construct isometric drawing of simple objects.	6 h	Use isometric snap/grid and isoplanes to draw cube, step block, cylinder-like objects.

Source information

Item	Status
Course code/title	2028 - Basic CAD Lab
Revision scheme	Revision 2021 site context; the supplied PDF page itself does not print a separate scheme label.

Credits/category/hours	No Credit; Engineering Science; 3 periods/week; 45 periods/semester.
CO-PO mapping	CO1 and CO2 moderately map to PO1 and PO4; CO3 and CO4 strongly map to PO1 and PO4.
Official model paper pattern	Not specified in supplied syllabus/source. Model paper here is a fresh practice paper for a practical CAD lab.

Clickable table of contents

Redesigned syllabus roadmap

Use this as your lab plan. Do not jump directly to drawing figures before setting units, layers, osnap and dimension style.

M1 • CO1 • 6 h

Core: CAD software, applications, advantages, hardware/software, AutoCAD window, new drawing.

Draw: software window diagram.

Explain: manual vs CAD; save/open/new drawing.

Common mistake: starting without units/limits.

M2 • CO2 • 9 h

Core: line, circle, arc, ellipse, rectangle, polygon, array, trim, mirror, offset, dimensions, text and plot.

Calculate: coordinate locations, scale and sheet size.

Common mistake: drawing by eye instead of coordinates/object snap.

M3 • CO3 • 21 h

Core: 2D entities, layers, hatch, orthographic views, sectional views, print/plot.

Draw: C-block, stepped block, cylinder block, lever, wall bracket.

Exam importance: highest scoring practical module.

M4 • CO4 • 6 h

Core: isometric concept, isometric snap/grid, isoplanes and simple objects.

Draw: cube, step block, cylinder-like object.

Common mistake: using normal circle instead of isocircle/ellipse method.

Learning roadmap table

Stage	Skill to master	What to produce	Inspection check
1	Setup	New drawing with units, limits, grid, snap, layers.	File saved with correct name, drawing area visible, layers named.

2	Geometry	2D shapes using exact command input.	No duplicate overlapping lines, endpoints closed, correct coordinates.
3	Editing	Trimmed, offset, mirrored, rotated and scaled geometry.	Edges meet cleanly; no broken corners.
4	Annotation	Dimensions, text, hatch and title information.	Readable, standard, non-overlapping dimensions.
5	Output	Print/plot on standard sheet.	Correct paper, scale, orientation, preview and line weights.

MODULE 1 • CO1 • 6 HOURS

Computer aided drafting software and CAD environment

Outcome: illustrate the use of CAD software, identify applications, advantages, components of the AutoCAD window and start a new drawing.

1.1 Meaning of CAD

Computer Aided Drafting is the use of computer software to create accurate technical drawings. In manual drafting, corrections require erasing and redrawing. In CAD, editing, copying, scaling, storing, plotting and reusing details are faster and more accurate.

> CAD □□□□□□ computer □□□□□□□□□□ accurate technical drawing □□□□□□□□□□□□□□□□□□□. Mouse □□□□□□ shape □□□□□□□□□□□□□□□ □□□□□□ □□□□□□; exact dimensions, coordinates, layer, plot scale □□□□□□ □□□□□□□□□□□□□□□□ □□□□□□□□.

1.2 Advantages over manual drafting

- Higher accuracy with coordinate input.
- Fast modification using trim, offset, mirror, array and scale.
- Reusable blocks for standard parts.
- Layer control for center lines, hidden lines, object lines and dimensions.
- Easy storage, sharing and printing.
- Better inspection traceability because files can be revised.

1.3 Hardware and software requirements

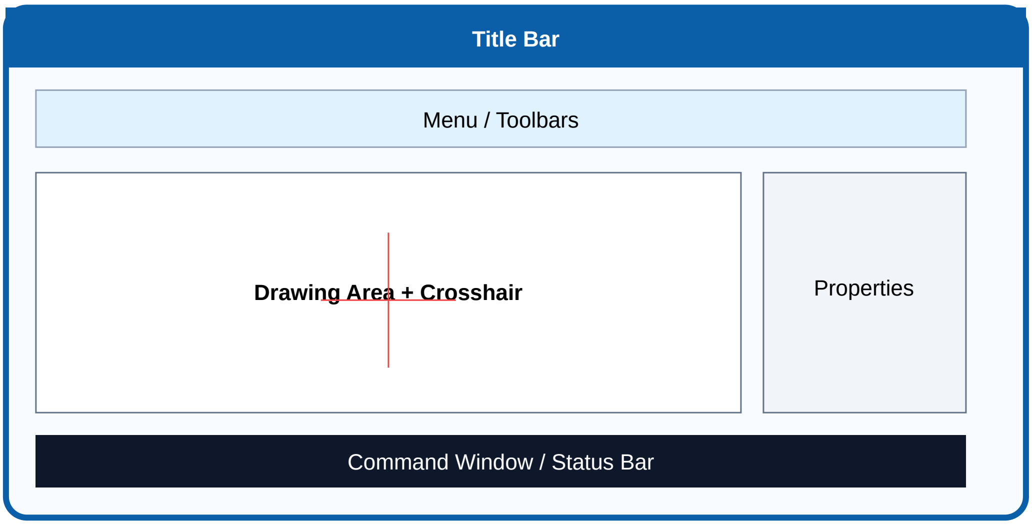
Hardware: computer/laptop, keyboard, mouse with scroll wheel, monitor, printer/plotter and storage device.
Software: AutoCAD or equivalent drafting tool, printer driver, PDF export tool and file backup location.

1.4 CAD software examples

The syllabus lists AutoCAD, Pro-E, IDEAS and open-source drafting software. AutoCAD is mainly used for 2D drafting practice. Pro-E/Creo type software is stronger for 3D modelling. Open-source CAD is useful for learning and low-cost drafting.

1.5 AutoCAD window components

Title bar, standard toolbar, menu bar, object properties toolbar, draw toolbar, modify toolbar, cursor crosshair, command window, status bar, drawing area and UCS icon form the basic working environment.



1.6 Starting and saving a drawing

1. Open CAD software and select new drawing/template.
2. Set units before drawing.
3. Set limits only if required by lab instruction.
4. Enable grid/snap when useful.
5. Save with course code, exercise number and student name.
6. Use Save As for revised versions instead of overwriting final record drawings.

Exam answer format: Define CAD, list four advantages, draw/label the software window, then explain new/open/save/save-as and command entry methods.

Module 1 expected questions

Define Computer Aided Drafting.

List advantages of CAD over manual drafting.

Name any four CAD software packages.

Draw and label an AutoCAD software window.

Explain command entry using mouse, pull-down menu and command window.

What are Units, Limits, Grid and Snap?

MODULE 2 • CO2 • 9 HOURS + LAB EXAM I

Commands, coordinates, dimensions, text and plotting

Outcome: identify and use core CAD commands, coordinate systems and plotting parameters.

Draw commands

LINE, CIRCLE, ARC, ELLIPSE, RECTANGLE, POLYGON, SPLINE and POLYLINE create geometry. Always watch the command line prompts.

Modify commands

ERASE, COPY, ARRAY, ROTATE, MIRROR, OFFSET, SCALE, MOVE, TRIM, FILLET, CHAMFER, EXTEND, STRETCH, PEDIT, EXPLODE, BLOCK and LAYERS control editing and reuse.

Precision tools

OSNAP, ORTHO, coordinates, direct distance entry, grid and snap prevent inaccurate freehand drawing.

Coordinate systems

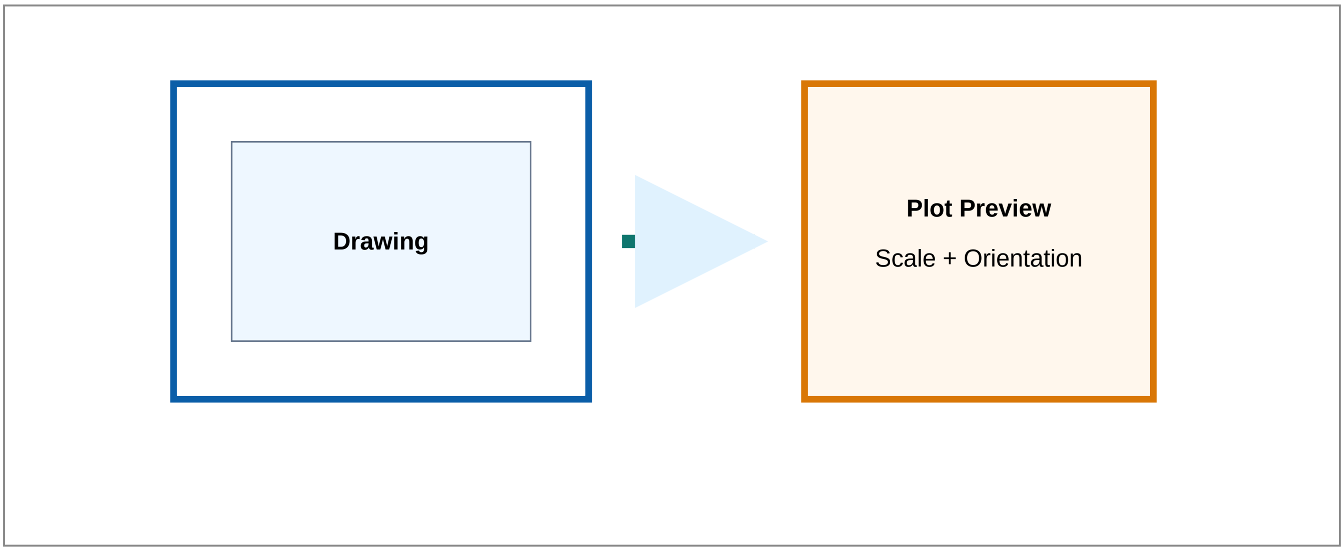
System	Input example	Meaning	Use
Absolute	100,50	Point measured from origin (0,0).	Locating fixed points.
Relative	@60,0	Point measured from previous point.	Rectangles, stepped profiles.
Polar	@80<30	Length and angle from previous point.	Inclined lines.
Direct distance	Move cursor direction + type 75	Length along cursor direction.	Fast orthogonal drawing with ORTHO.

Dimensioning and text

Dimension commands include linear, horizontal, vertical, aligned, rotated, baseline, continuous, diameter, radius and angular dimensions. Text may be single-line or multiline. Dimension style must be readable; tiny text or crowded dimensions lose marks.

Plotting parameters

Before printing, select paper size, paper units, orientation, plot area, plot scale, plot offset, line weight and print preview. Never print without preview because wrong scale wastes record sheets.



Common lab failure: using mouse-only drawing without ORTHO/OSNAP/coordinates. It may look close on screen, but dimensions will expose the error.

Module 2 expected questions

- Explain absolute, relative and polar coordinates with examples.
- List five draw commands and five modify commands.
- Differentiate TRIM and EXTEND.
- Explain baseline and continuous dimensions.
- Write plotting parameters required before printing.
- Draw a rectangle using relative coordinate method.

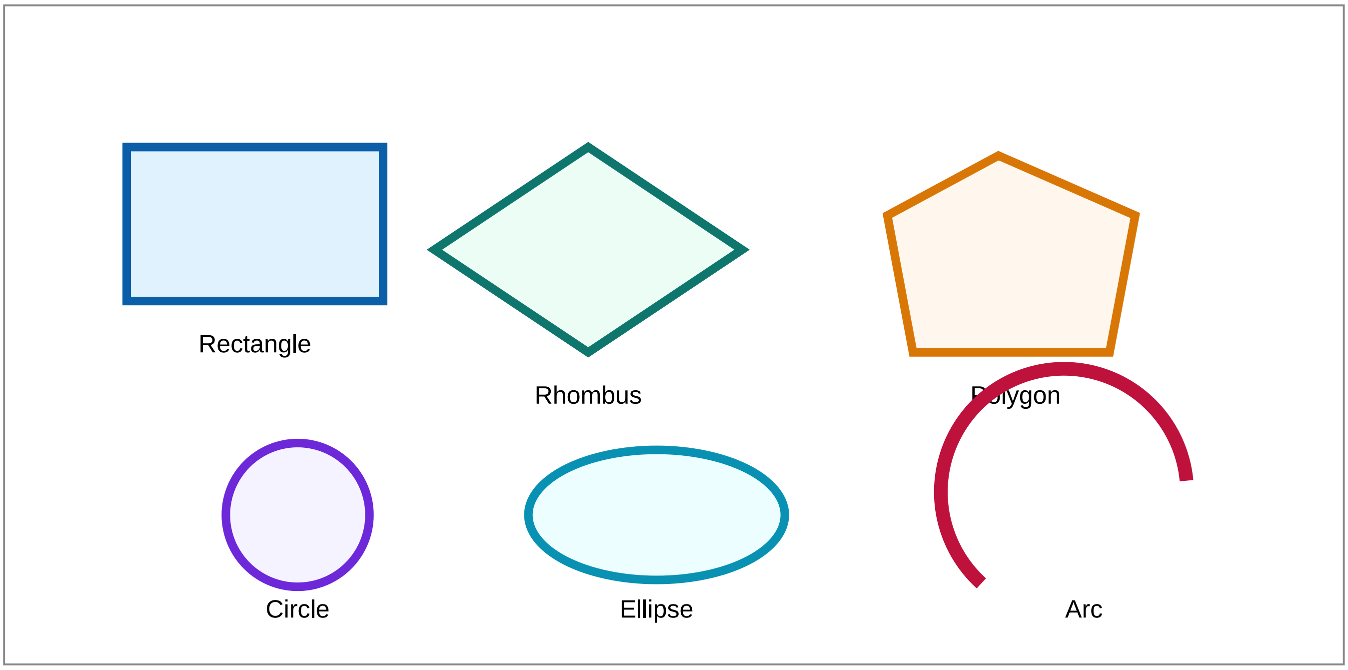
MODULE 3 • CO3 • 21 HOURS

2D entities, orthographic views, sectional views and plotted output

This is the largest scoring module. It requires real drawing practice, not reading only.

3.1 Basic 2D entities

Practice starts with rectangle, rhombus, polygon, circle, arc, ellipse, rectangular array, circular array and blocks. Each entity must be drawn with exact dimensions.



3.2 Layers and line properties

Use separate layers for object line, hidden line, center line, dimension, hatch and construction. Assign line type, line weight and color systematically. Layer discipline is what makes CAD drawing professional.

Layer	Line type	Use
Object	Continuous thick	Visible edges
Hidden	Dashed	Invisible edges
Center	Center line	Axis, holes, symmetry
Dimension	Continuous thin	Dimension and extension lines

3.3 Orthographic views

Orthographic drawing represents front, top and side views on 2D planes. The front view should show maximum shape information. Projection alignment must be maintained; do not randomly place views.

> Orthographic view-□ □□□ object-□□ front/top/side □□□ □□□□□□□□□□□ □□□□□□□□□□□□. Views □□□□□□ alignment □□□□□□□□□□ drawing technically wrong □□□.

3.4 Sectional views

Sectional view shows the interior of a component by imagining a cutting plane. Hatching must be uniform and should not hide dimensions or center lines. Use sections for lever, wall bracket and internal features.

3.5 Required exercises

- Minimum 5 orthographic drawings such as C-block, stepped block and cylindrical block.
- Minimum 3 sectional drawings such as lever and wall bracket.
- Print/plot prepared drawings and keep them in record book.

3.6 Plot record quality

Record printout should include correct drawing number, title, scale, name, date, border/title block if instructed, readable dimensions and no missing geometry.

Module 3 expected questions

Draw basic 2D entities using CAD commands.

Explain layer properties and line weight.

Draw orthographic views of a simple stepped block.

Draw sectional view of a wall bracket.

Explain hatching and gradient use.

Write the plotting procedure for a prepared drawing.

MODULE 4 • CO4 • 6 HOURS + LAB EXAM II

Isometric drawing of simple objects

Outcome: understand isometric concept and construct simple isometric drawings using isometric snap/grid.

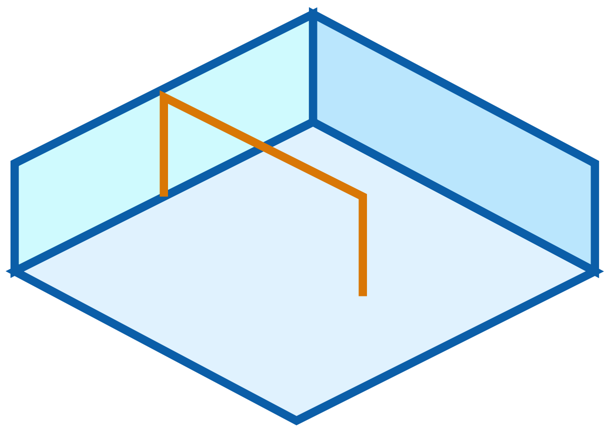
4.1 Isometric concept

Isometric drawing represents a 3D object on a 2D sheet using three principal axes. The axes are usually vertical and 30 degrees to the horizontal on left and right sides. Isometric drawing shows shape clearly but is not a perspective drawing.

4.2 Isometric snap and grid

Isometric snap changes cursor movement to isometric directions. Isoplanes help draw left, right and top planes. Use F5 or equivalent option to switch isoplane in CAD software.

Isometric step block diagram



Isometric view uses 30° axes and vertical heights

Procedure for isometric drawing

1. Set units and open isometric snap/grid.
2. Choose starting corner and draw the base using isometric axes.
3. Project vertical heights from corners.
4. Complete visible edges with continuous line.
5. Use isocircle/ellipse method for circular faces.
6. Add dimensions parallel to isometric axes.

Common mistake: drawing an isometric circle as a normal circle. In isometric view, a circular face appears as an ellipse/isocircle.

Module 4 expected questions

- Define isometric drawing.
- Explain isometric snap and grid.
- Draw isometric view of a cube.
- Draw isometric view of a step block.
- Explain how to represent a circular face in isometric drawing.

CAD command, rule and procedure bank

LINE

Use: straight segments. **Example:** LINE → 0,0 → @100,0. **Rule:** use ORTHO for horizontal/vertical lines.

OFFSET

Use: parallel copy at exact distance. **Example:** OFFSET → 10 → select object → side. Used for wall thickness, slots and profiles.

TRIM

Use: remove extra portions up to cutting edges. Check corners after trimming; do not leave small gaps.

MIRROR

Use: symmetrical drawings. Draw half accurately, mirror about center line and check dimensions.

ARRAY

Use: repeated objects. Rectangular array for rows/columns; polar array for circular repetition.

HATCH

Use: sectioned material. Keep hatch angle and spacing readable.

Practical checklist before submitting a CAD record drawing

- Units set correctly before drawing.
- All geometry made using exact dimensions or coordinates.
- Layer names and line types are suitable.
- OSNAP used for endpoint, midpoint, center, intersection and tangent where needed.
- Dimensions are readable and not overlapping.
- Title block and drawing name are complete.
- Print preview checked before plotting.

Solved examples / problem lab

Example 1 - rectangle using relative coordinates

Problem: Draw a 100 mm x 60 mm rectangle starting from origin.

1. Command: LINE.
2. First point: 0,0.
3. Next points: @100,0 → @0,60 → @-100,0 → C.
4. Check dimension: horizontal 100, vertical 60.

Final: closed rectangle of 100 x 60 mm.

Example 2 - concentric circles

Problem: Draw two circles of radius 25 mm and 40 mm with same center.

1. Command: CIRCLE.
2. Center: 100,100; radius: 25.
3. Repeat CIRCLE; center: 100,100; radius: 40.
4. Use center osnap if selecting center from first circle.

Exam tip: never guess center by eye.

Example 3 - offset slot

Problem: Draw a 20 mm wide slot from a center line.

1. Draw center line.
2. OFFSET 10 mm both sides.
3. Add semicircular ends using circle/arc.
4. TRIM unwanted portions.

Example 4 - orthographic answer planning

Given: simple stepped block.

1. Select front view showing most details.
2. Project top view vertically from front view.
3. Project side view horizontally.
4. Add hidden/center lines where required.
5. Dimension after geometry is complete.

Example 5 - plotting procedure

1. Open layout or plot command.
2. Select printer/PDF plotter.
3. Select A4 or A3 as instructed.
4. Choose portrait/landscape.
5. Select plot area/window.
6. Set scale and center plot.
7. Preview, correct errors, then print.

Practice section

Objective questions

1. Which command creates parallel copies? A) Trim B) Offset C) Erase D) Explode
2. Which coordinate input means 80 mm at 30 degrees from previous point? A) 80,30 B) @80<30 C) @30<80 D) 80@30
3. Which tool helps pick exact endpoint? A) OSNAP B) Hatch C) Plot D) Save As
4. Which command repeats objects in rows/columns? A) Array B) Fillet C) Stretch D) Pedit
5. In sectional view, cut material is usually shown by A) Hatch B) Mirror C) Grid D) Zoom
6. Before printing, which option is essential? A) Explode B) Preview C) Undo D) Arc

Drawing practice tasks

- Draw a rectangle, rhombus, polygon, circle, arc and ellipse with dimensions.
- Create rectangular and circular array exercises.
- Draw orthographic views of C-block, stepped block and cylindrical block.
- Draw sectional view of lever and wall bracket.
- Draw isometric view of cube, step block and cylinder-like object.

Answer key

1-B, 2-B, 3-A, 4-A, 5-A, 6-B.

Practical checking: Answer is correct only when drawing dimensions and plot output are correct.

Fresh model practical question paper

This is a practice model because the supplied syllabus does not specify an official written model question paper pattern.

Course Code: 2028
Course Title: Basic CAD Lab
Time: 3 Hours
Maximum Marks: Practice pattern

Part A - CAD setup and commands
1. Start a new drawing, set units, limits, grid and snap. Create layers for object, hidden, center, dimension and hatch.
2. Draw a labelled CAD software window showing title bar, menu/toolbars, drawing area, command window, status bar and UCS icon.

Part B - 2D drafting task
3. Draw a 120 mm x 70 mm base rectangle. Add a 40 mm diameter circle at the center. Add two 20 mm diameter holes symmetrically. Use center lines and dimensions.
4. Create an offset profile of 10 mm inside the rectangle and trim unwanted portions.

Part C - Orthographic / sectional drawing
5. Draw front, top and right side views of a simple stepped block using proper projection alignment and dimensions.
OR
6. Draw sectional front view of a wall bracket with hatch, hidden lines where required and dimensions.

Part D - Isometric drawing and plotting
7. Draw an isometric view of a simple step block using isometric snap/grid and add dimensions.
8. Plot any one completed drawing on A4 sheet/PDF with correct orientation, scale, line weight and preview.

Complete model answer guide

Answer 1 - setup

Set units according to lab instruction, apply limits if required, enable grid/snap if useful, create named layers and assign proper linetype/lineweight. Save the file before drawing.

Answer 2 - CAD window

Draw a rectangular software interface and label title bar, standard toolbar, menu bar, draw toolbar, modify toolbar, object properties toolbar, drawing area, cursor crosshair, command window, status bar and UCS icon.

Answer 3 - 2D drafting

Use rectangle with exact coordinates, place center circle using CIRCLE command, locate hole centers symmetrically using construction/center lines or coordinates, draw holes, add center marks and dimensions. Do not place holes by eye.

Answer 4 - offset profile

Use OFFSET 10, select rectangle, choose inside, then TRIM corners/extra portions as required. Check the inside profile is closed.

Answer 5/6 - orthographic or section

Select correct front view, project top and side views, use line types correctly, add dimensions after construction. For sectional view, show cutting plane result, hatch cut surfaces uniformly and avoid hatching holes/voids.

Answer 7 - isometric

Switch isometric snap/grid, select correct isoplane, draw base, project vertical heights, complete visible edges and use ellipse/isocircle for circular faces. Dimensions must follow isometric direction.

Answer 8 - plot

Select printer/PDF plotter, A4 paper, orientation, plot area/window, plot scale, center plot and preview. Correct any missing lineweight, scale or clipping before printing.

Common mistakes to avoid

- Drawing without setting units.
- Using freehand mouse positions instead of coordinates or OSNAP.
- Wrong layer/line type for hidden and center lines.
- Overlapping dimensions and unreadable text.
- Normal circle used in isometric drawing.
- Printing without preview.

References

Type	Reference
Official source	Supplied official syllabus PDF for Course 2028 - Basic CAD Lab.
T1	P I Varghese, K C John, Engineering Graphics, VIP Publishers.
R1	N. D. Bhatt, Engineering Drawing, Charotar Publishing House, Anand, Gujarat, 2010; ISBN 978-93-80358-17-8.
R2	Kulkarni, D. M.; Rastogi, A. P.; Sarkar, A. K., Engineering Graphics with AutoCAD, PHI Learning Private Limited, New Delhi, 2010. ISBN number is incomplete/truncated in supplied PDF.
R3	Jeyapoovan, T., Essentials of Engineering Drawing and Graphics using AutoCAD, Vikas Publishing House Pvt. Ltd., Noida, 2011; ISBN 978-8125953005.
R4	Autodesk, AutoCAD User Guide, Autodesk Press, USA, 2015.
R5	Sham Tickoo, AutoCAD 2016 for Engineers and Designers, Dreamtech Press / Galgotia Publication, New Delhi, 2015; ISBN 978-9351199113.
Online	www.swayam.gov.in; https://nptel.ac.in/course.html; https://ndl.iitkgp.ac.in/